SCX Research Supercomputing Configuration Checklist

Wallfacer R&D Workstation · 2026-06-30

I. Compute Layer

Component	Configuration	Purpose	Budget
Main GPU	1× RTX 4090 24GB (existing)	NEP training, small model inference, Spring self-evolution	—
Inference GPU Pool	4× RTX 4090 24GB	Parallel CC sessions, multi-expert voting, M-seed decoding	¥50,000
Supercomputer	Xiaogan Supercomputing Center (existing)	DFT computation, large-scale database Full Screening	—
CPU	AMD Threadripper 7970X (32 cores)	Data preprocessing, feature extraction	¥15,000
Memory	128GB DDR5 ECC	200GB drug database memory mapping	¥4,000
Storage	2× 4TB NVMe SSD (RAID0)	Database high-speed read/write, CC workspace	¥6,000

Component	Configuration	Purpose	Budget
Storage	1× 20TB HDD	Cold data archiving, DFT trajectories	¥3,000
Network	10 Gigabit fiber	Supercomputer data transfer	—

II. AI Engine Layer

Engine	Model	Purpose	Monthly Cost
CC Main	DeepSeek V4 Pro (anthropic endpoint)	Math derivation, paper writing, code generation	¥500
CC Review	DeepSeek V4 Pro × 3 parallel	Multi-expert hostile review	¥1,500
Local Inference	Qwen-7B / DeepSeek-7B (4090)	Rapid prototyping, M-seed decoding	Free
API Backup	Claude Sonnet 4 (OpenRouter)	Fallback when DeepSeek overloaded	¥200
Embedding Model	BGE-M3 (local)	State Crystallization feature clustering	Free

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III. Software Stack

Layer	Tool	Configuration
OS	Ubuntu 24.04 LTS	WSL2 or bare metal
Shell	zsh + tmux	Multi-CC session management
Python	3.11 + uv	Project dependency management

Layer	Tool	Configuration
LaTeX	TeX Live 2025	Paper compilation
Git	main + feature branches	SCX version management
Editor	VS Code + LaTeX Workshop	Paper writing
Container	Docker + NVIDIA Container Toolkit	Environment isolation

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IV. CC Workflow Optimization

Config Item	Recommended Value	Rationale
`max-turns`	50 (math derivation) / 30 (paper editing)	Complex proofs need more turns
`effort`	max (derivation) / high (review)	Math needs deep reasoning
Parallel CC count	3-4 way	Supported by RTX 4090 $\times$ 4
Single-way timeout	600s	50 turns max effort typically 3-8 min
`--allowedTools`	Read,Write,Edit (papers); Read,Write,Edit,Bash (code)	Avoid accidents
Working Directory	`/g/Xiaogan_Supercomputing_data/SCX`	Unified path
Full path notation	`G:/Xiaogan_Supercomputing_data/SCX/`	CC workspace on D: drive, needs absolute path

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V. Data Layer

Dataset	Size	Purpose	Status
AlN v3 (DFT)	534 frames	SCX validation	Existing
DrugBank 5.0	6,784 entries	Yajie drug audit	To download

Dataset	Size	Purpose	Status
ChEMBL 34	~2M entries	Full Screening	To download
PubChem	~100M entries	Full Screening	To download
BindingDB	~2M entries	Full Screening	To download
PDBbind	~20K entries	Full Screening	To download
Other 7 databases	~50GB	Full Screening	To download
Drug Total	**~200GB**	**12 databases full scale**	

VI. Security Layer

Measure	Configuration
Git remote backup	GitHub private repo
Cold backup	External HDD, weekly rsync
`.env` protection	`.gitignore` excludes API key
Paper versioning	Commit after each hostile review round
Theory files	Archived under `theory/` by date

VII. Shopping List (Priority)

Priority	Device	Budget	Rationale
P0	4× RTX 4090	¥50,000	Parallel CC + NEP training
P1	4TB NVMe SSD × 2	¥6,000	200GB database + workspace
P2	Threadripper + 128GB	¥19,000	Data preprocessing
P3	20TB HDD	¥3,000	Cold backup
Total		**¥78,000**	

VIII. What Not to Buy

Item	Rationale
A100/H100	Not needed. DeepSeek API + 4090 suffices for all derivations
Cloud Computing	Supercomputer already available. API monthly ¥2,000 suffices
Mac Studio	Ecosystem incompatible with CUDA
Multi-node Cluster	Not needed for single-person research